

# Assignment 7 Report: Sound Localization

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## Abstract

This report presents the implementation of the Delay-and-Sum (DAS) algorithm to reconstruct ultrasound images by estimating the position of obstacles. The algorithm involves simulating an array of microphones that receive reflected sound waves from obstacles and using these delayed signals to form a composite image. Experiments were conducted with varying numbers of microphones and sample counts to analyze the effects on reconstruction quality.

## 1 Introduction

The goal of this assignment is to implement the DAS algorithm, commonly used in ultrasound imaging, to reconstruct images by estimating the position of obstacles. This approach uses an array of microphones, each receiving delayed sound signals, and applies beamforming to estimate the obstacle's location.

## 2 Delay-and-Sum Algorithm (DAS)

The DAS algorithm is a beamforming technique that focuses on a specific direction of arrival by delaying and summing the outputs of individual sensors. Each microphone receives a delayed version of the sound wave based on its distance from the source. These signals are then summed after correcting for their respective delays to focus on a particular direction.

### 2.1 Assumptions

The following assumptions were made:

- The signal received by each microphone is a delayed version of the source wave.
- Each microphone's received wave is symmetric around its delay time.
- The received signal has an impulse at  $t = 0$ , matching the source's impulse.

## 3 Procedure

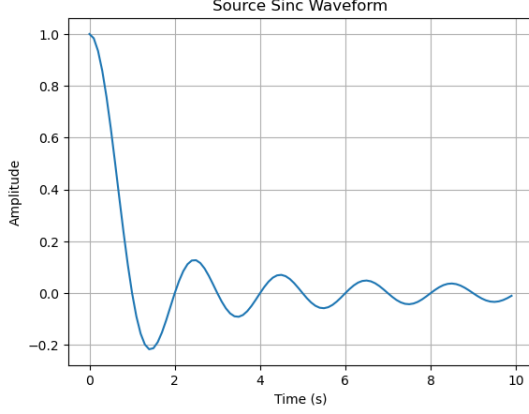
1. Microphone locations were defined symmetrically along the Y-axis at  $x = 0$  and spaced by a defined pitch distance. Microphone placement considered whether the number of microphones  $N_{\text{mics}}$  was even or odd, ensuring that no microphone was positioned exactly at the origin.
2. The source sound wave was generated using a sinc function, with time sampling determined by the speed of sound  $C$ , the distance per sample, and a scaling parameter  $\text{SincP}$ . The sinc pulse's narrowness was adjusted with  $\text{SincP}$ , affecting the resolution of the reconstructed image.
3. The time delay for each microphone was calculated based on the distances from the source to the obstacle and from the obstacle to each microphone. This approach helped account for the signal's journey to each receiver after reflection.
4. The DAS algorithm was applied by summing the signals received by each microphone, shifted by their respective time delays. This step was performed over a 2D grid to simulate the reconstruction process.
5. A heatmap of the summed signals was generated, representing the estimated position of the obstacle within the grid based on the signals' alignments.

## 4 Results and Observations

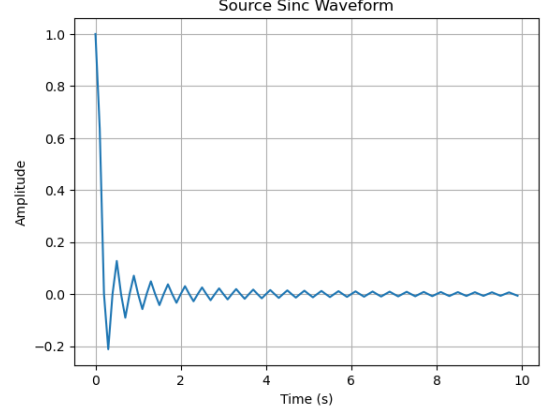
### 4.1 Step 1: Generating and Analyzing Sinc Pulses

The source sound wave was generated using a sinc pulse, where the  $SincP$  parameter directly controlled the width of the pulse. Adjusting  $SincP$  allowed for finer or coarser image resolution:

- A lower  $SincP$  generated smoother, broader pulses that reduced detail but increased smoothness in the image.
- A higher  $SincP$  produced sharper, more detailed images by compressing the pulse width, leading to higher resolution.



(a) Sinc function with  $SincP = 1$ .

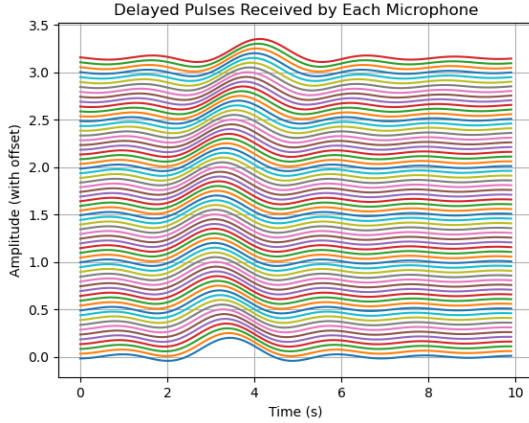


(b) Sinc function with  $SincP = 5$ .

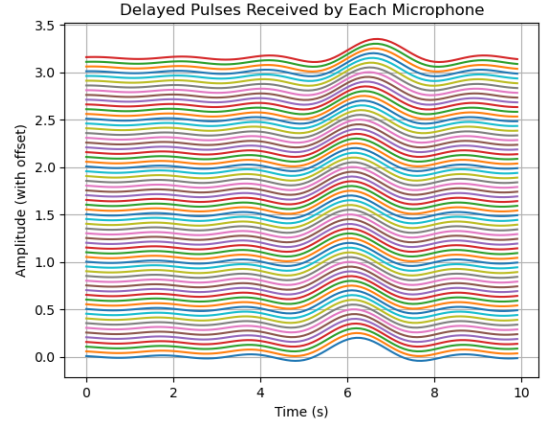
Figure 1: Plots of sinc functions with varying  $SincP$  values.

### 4.2 Step 2: Changing Obstacle Locations

Experiments were conducted with obstacle positions at  $(3, -1)$  and  $(6, -1)$ , analyzing their impact on the reconstructed image.



(a) Obstacle at  $(3, -1)$

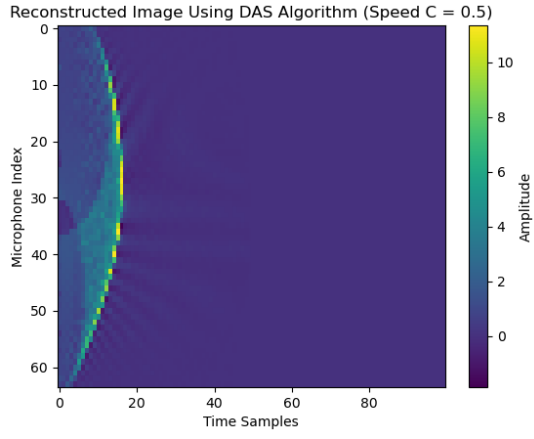


(b) Obstacle at  $(6, -1)$

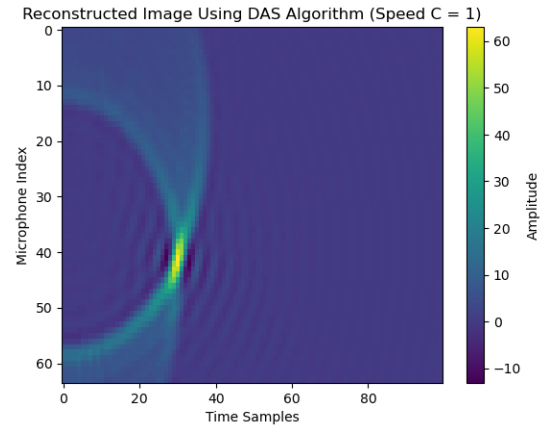
Figure 2: Reconstructed images for different obstacle locations.

### 4.3 Step 4: 9 Combinations of $N_{\text{mics}}$ and $N_{\text{samp}}$

Below are sample outputs for different combinations of  $N_{\text{mics}}$  and  $N_{\text{samp}}$ .



(a)  $C = 0.5$



(b)  $C = 1$

Figure 3: Reconstructed images with different values of speed of sound  $C$ .

#### 4.4 Data Files RX2 and RX3

### 5 Conclusions

Using 64 microphones provides the best vertical range for accurate reconstruction, while higher sample rates improve clarity. Adjusting parameters such as the sinc pulse width  $SincP$  and speed of sound  $C$  affects resolution and sharpness in the reconstructed image. The DAS algorithm, applied under appropriate configurations, effectively estimates the position of obstacles in ultrasound imaging.

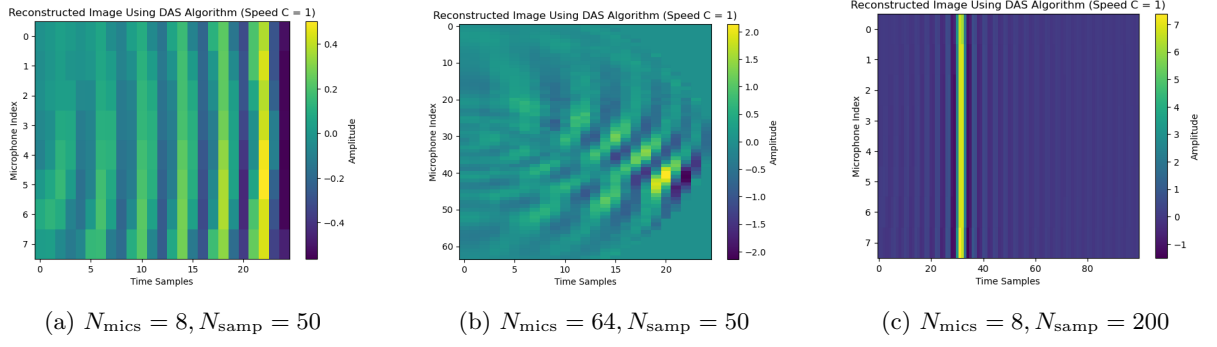


Figure 4: Reconstructed images with various combinations of  $N_{\text{mics}}$  and  $N_{\text{samp}}$ .

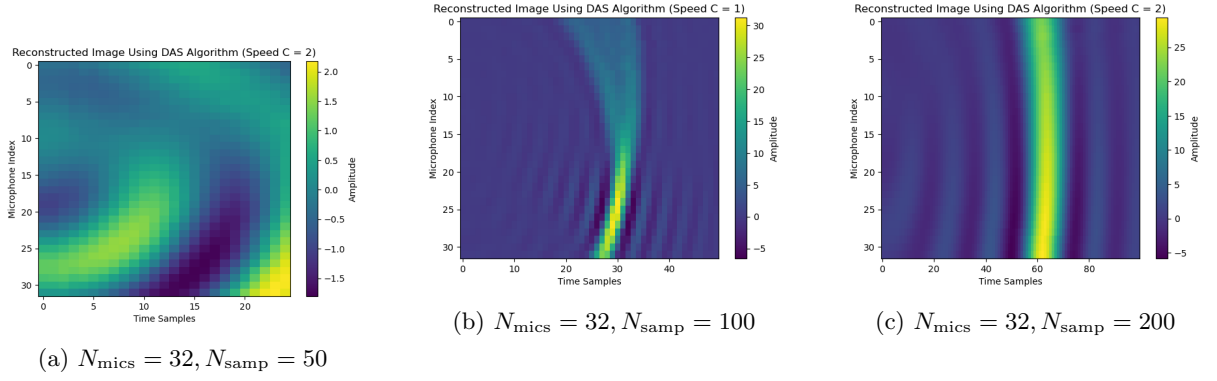


Figure 5: Reconstructed images with various combinations of  $N_{\text{mics}}$  and  $N_{\text{samp}}$ .

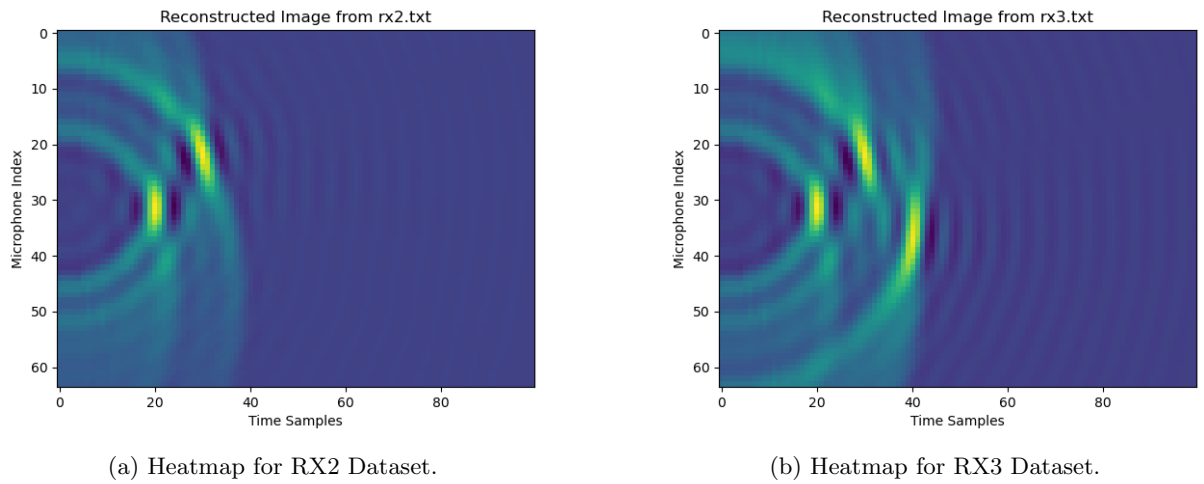


Figure 6: Heatmaps of the RX2 and RX3 Datasets.